Symmetry in Exact diagonalization method

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The center of many-body physics is to diagonalize the Hamiltonian $\mathcal{H}$ in the Hilbert space $\mathscr{H}$. To be explicit, we have

$$\mathscr{H} = \text{span}\{|\sigma_1 \dots \sigma_L\rangle \mid \sigma_i = 1, \dots, d\} \tag{1}$$

The problem with this approach is that the dimension of $\mathscr{H}$ is exponentially large. Let L be the lattice size and d be the local Hilbert space dimension, the dimension of $\mathscr{H}$ is d^L. If the system has $\mathsf{U}(1)$ symmetry, then $\mathscr{H}$ is further decomposed into subspaces $\mathscr{H}_N$ with fixed particle number N, defined as

$$\mathscr{H}_N = \text{span}\{|\sigma_1 \dots \sigma_L\rangle \mid \sum_{i=1}^{L} \sigma_i = N\}, \tag{2}$$

and the dimension of $\mathscr{H}_N$ is $\binom{L}{N}$, which is still exponentially large.

To address such problem, we will use the representation theory[1] to decompose the Hilbert space into irreducible representations of the symmetry group G of the Hamiltonian $\mathcal{H}$. We will assume G is a finite group, and it acts on the Hilbert space $\mathscr{H}$ via a unitary representation $U : G \to U(\mathscr{H})$.

1 Building the symmetry group

In our case case, we consider two kinds of symmetry: the lattice symmetry that permutes the sites and the internal symmetry that performs a spin flip:

$$g\,|\sigma_1 \dots \sigma_L\rangle = \left|\sigma_{g^{-1}(1)} \dots \sigma_{g^{-1}(L)}\right\rangle, \quad g \in G_{\text{latt}} \tag{3}$$

$$\mathcal{F}\,|\sigma_1 \dots \sigma_L\rangle = |(d-\sigma_1), \dots, (d-\sigma_L)\rangle \tag{4}$$

and the full symmetry group is $G = G_{\text{latt}} \times \mathbb{Z}_2$. The lattice symmetry group G_{latt} is further decomposed into a semi-direct product of the translation group G_{tr} and the point group G_{pt}, i.e. $G_{\text{latt}} = G_{\text{tr}} \rtimes G_{\text{pt}}$, where $G_{\text{tr}} = \mathbb{Z}_{N_1} \times \cdots \mathbb{Z}_{N_d}$ is a finite Abelian group and G_{pt} is a finite group.

Suppose we already know all the information of G_{pt}, we need to

1. Find the conjugacy classes of G_{latt} and G.
2. Calculate the character table of G_{latt} and G.

1.1 The lattice Group

First we describe the structure of the lattice group $G_{\text{latt}} = G_{\text{tr}} \rtimes G_{\text{pt}}$. Let $t \in G_{\text{tr}}$ and $r \in G_{\text{pt}}$. The element in G_{latt} can be written as (t, r), with the group multiplication defined as

$$(t_1, r_1)(t_2, r_2) = (t_1 + r_1 t_2, r_1 r_2) \tag{5}$$

where $r(t)$ is the action of r on t. The inverse of (t, r) is given by

$$(t, r)^{-1} = (-r^{-1}t, r^{-1}) \tag{6}$$

It acts on the Hilbert space as

$$(t, r) \left|\sigma_1 \ldots \sigma_L\right\rangle = \left|\sigma_{r^{-1}(t^{-1}(1))} \ldots \sigma_{r^{-1}(t^{-1}(L))}\right\rangle \tag{7}$$

Conjugacy Class Since G_{tr} is a normal subgroup of G_{latt}, the conjugacy class of G_{latt} is divided into two cases:

1. Since

$$\begin{aligned} (a, r)(t, 1)(a, r)^{-1} &= (a + rt, r)(-r^{-1}a, r^{-1}) & (8) \\ &= (a + rt - r^{-1}a, 1) & (9) \end{aligned}$$

for $(t, 1)$ and $(t', 1)$ to be in the same conjugacy class, we need $t' = a + rt - r^{-1}a$ for some $a \in G_{\text{tr}}$ and $r \in G_{\text{pt}}$.

2. Translations and point group elements. We have

$$\begin{aligned} (a, r)(t, s)(a, r)^{-1} &= (a + rt, rs)(-r^{-1}a, r^{-1}) & (10) \\ &= (a + rt - r^{-1}a, rsr^{-1}) & (11) \end{aligned}$$

Hence for (t, s) and (t', s') to be in the same conjugacy class, we need s and s' to be in the same conjugacy class of G_{pt}, and $t' = a + rt - r^{-1}a$ for some $a \in G_{\text{tr}}$ and $r \in G_{\text{pt}}$.

Character The irreducible representations of G_{tr} are one-dimensional, and they are labeled by the momentum

$$k \in \mathrm{BZ} = \left\{ (k_1, \ldots, k_d) \mid k_i = 0, \frac{2\pi}{N_i}, \ldots, \frac{2\pi(N_i - 1)}{N_i} \right\}, \tag{12}$$

with the character defined as

$$\rho_k(t) = e^{ik \cdot t}. \tag{13}$$

The irreducible representations of G_{latt} are labeled by the momentum k and the irreducible representations of the little group $G_k = \{r \in G_{\mathrm{pt}} \mid rk = k\}$,

References

[1] William Fulton and Joe Harris. *Representation Theory: A First Course.* Springer, 1991.